

Superposition

More features than dimensions · video 2:07

This lesson answers one question: why do single neurons in language models respond to many unrelated things? A toy model gives the answer: when features are **sparse** (rarely active), a network can store more of them than it has dimensions, in overlapping directions: **superposition**. Five features squeezed into two dimensions go from two (dense) to all five (97% sparse, a pentagon); a hundred features in twenty dimensions go from 14 to all 100. The price is neurons that are **polysemantic**.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 6).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. The toy model	▶ 0:08–0:47	1	☐
2. Sparsity enables superposition	▶ 0:47–1:35	3	☐
3. What it means for real models	▶ 1:35–1:59	1	☐

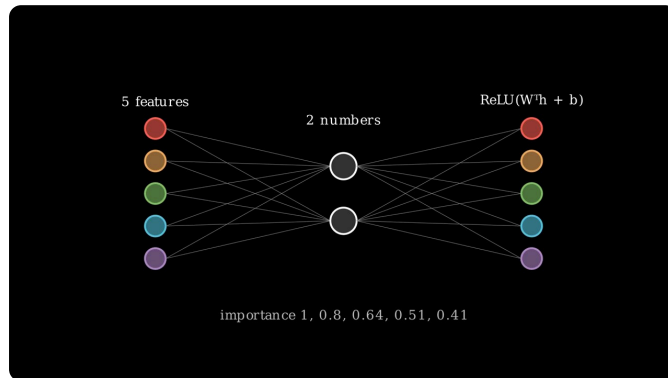
Before you start: Deep Dive episodes 15 (activations) and 13 (the residual stream). Code: `code/d43_superposition` (PyTorch only, a few minutes on a CPU).

CONCEPT 1 OF 3

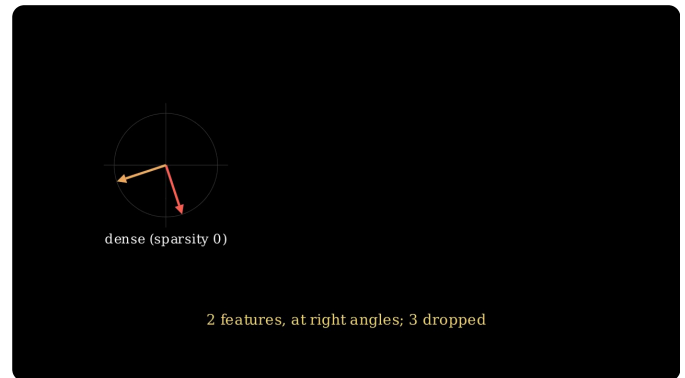
The toy model

▶ Video 0:08-0:47

The toy model compresses n features into $m < n$ numbers, $h = Wx$, and tries to read them back: $x' = \text{ReLU}(W^T h + b)$. Each column of W is a feature's direction in the small space. Each feature is zero with probability S (the **sparsity**), otherwise uniform in $[0, 1]$; the loss weights feature i by its importance, here 0.8^i .



▶ 0:36 5 features $\rightarrow$ 2 numbers $\rightarrow \text{ReLU}(W^T h + b) \rightarrow$ 5 features; feature i has importance 0.8^i .



▶ 0:45 Dense features: the two most important get perpendicular directions; three are dropped.

With dense features ($S = 0$) and 5 features in 2 dimensions, the model keeps the two most important, with directions of length 1.00 at right angles, and drops the other three (lengths 0.00-0.02): exactly what PCA would do.

KEY IDEA

Without sparsity, m dimensions hold m features; the rest are lost.

Check 1: The toy model

Answer all 1 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

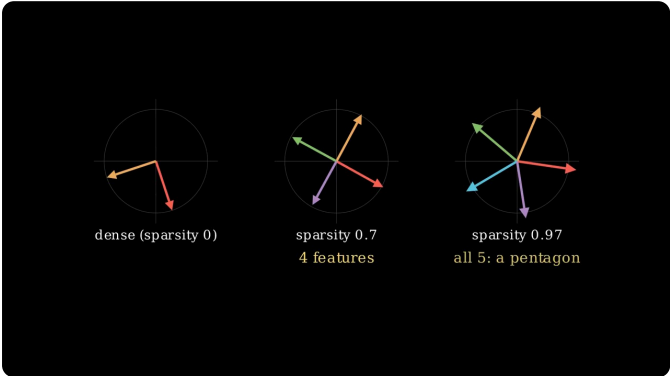
With $S = 0.97$ and 5 features, how many features are active in an average example?

.....

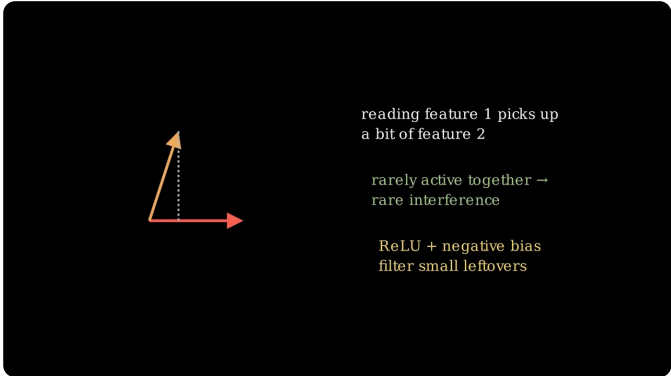
Sparsity enables superposition

▶ Video 0:47-1:35

Make the features rare and the model packs more of them: 4 features at $S = 0.7$, all 5 at $S = 0.97$, spread about 72° apart (angles $-149^\circ, -81^\circ, -8^\circ, 67^\circ, 140^\circ$): a pentagon. The directions overlap, so reading one feature picks up a bit of the others; but if features are rarely active together, that interference is rare, and the ReLU with a negative bias filters out the small leftovers.



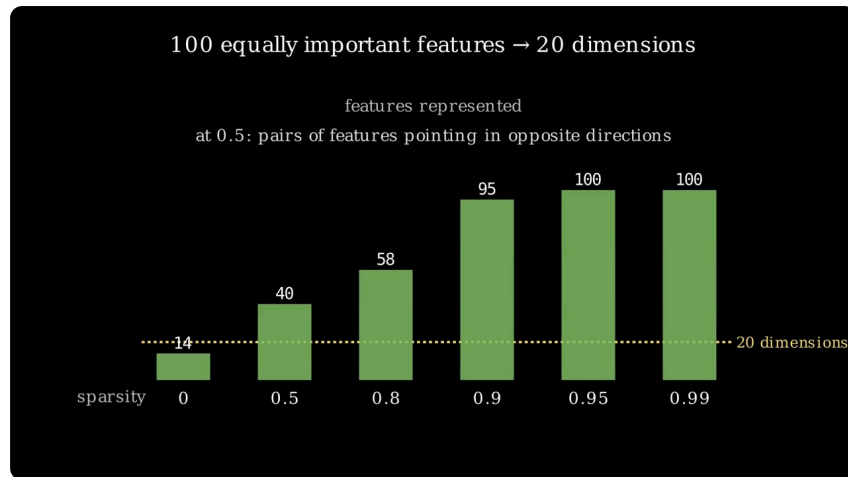
▶ 1:01 Directions of the 5 features: dense (2), sparsity 0.7 (4), sparsity 0.97 (all 5, about 72° apart).



▶ 1:15 Overlapping directions cause interference; rare co-activation plus ReLU and a negative bias keep it small.

SPARSITY	FEATURES REPRESENTED (OF 100, IN 20 DIMS)	SIGNED COSINE WITH CLOSEST OTHER FEATURE
0	14	-0.18
0.5	40	-1.00 (opposite pairs)
0.8	58	-0.80
0.9	95	-0.50
0.95	100	-0.48
0.99	100	-0.49

At sparsity 0.5 the model uses **antipodal pairs**: two features share one direction with opposite signs, so the ReLU separates them. From 0.95 on, all 100 features fit: five per dimension.



▶ 1:33 100 equally important features in 20 dimensions: 14, 40, 58, 95, 100, 100 represented at sparsity 0, 0.5, 0.8, 0.9, 0.95, 0.99.

KEY IDEA

The sparser the features, the more of them fit in superposition, at the cost of interference.

Check 2: Sparsity enables superposition

Answer all 3 questions before looking at the answers. Pass mark: all correct.

2.1 CALCULATION

Five unit vectors evenly spaced in a plane: what is the angle between neighbours, and their cosine?

2.2 SHORT ANSWER

Why can two features share one direction with opposite signs (cosine -1)?

2.3 HANDS-ON CODE

Try it yourself. Train the $5 \rightarrow 2$ model at $S = 0.97$ and print the angles.

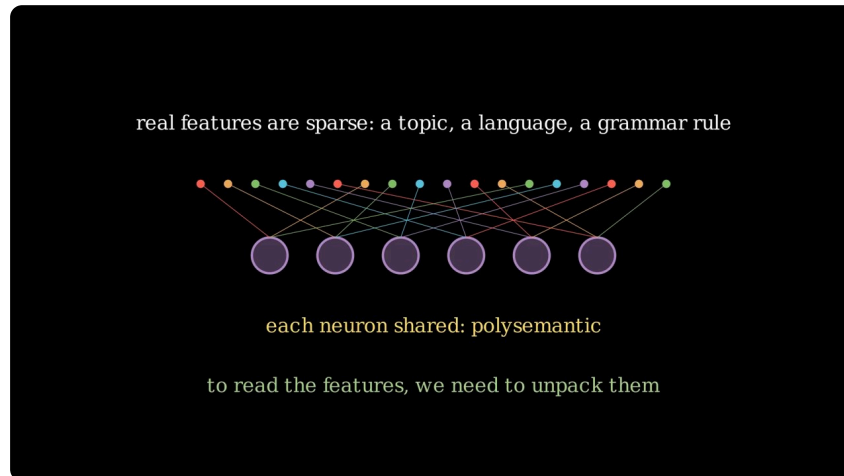
```
import torch
torch.manual_seed(0)
W = torch.nn.Parameter(torch.randn(2, 5) * 0.1); b = torch.nn.Parameter(torch.zeros(2))
imp = 0.8 ** torch.arange(5.); opt = torch.optim.Adam([W, b], lr=1e-2)
for _ in range(6000):
    x = torch.rand(1024, 5) * (torch.rand(1024, 5) > 0.97)
    loss = (imp * (torch.relu(x @ W.T @ W + b) - x) ** 2).mean()
    opt.zero_grad(); loss.backward(); opt.step()
print(W.norm(dim=0), torch.atan2(W[1], W[0]) * 180 / torch.pi)
```

CONCEPT 3 OF 3

What it means for real models

▶ Video 1:35-1:59

Real features (a topic, a language, a grammar rule, a named entity) are sparse: each is active on a small fraction of tokens. So large models likely store many more features than they have neurons, and each neuron ends up responding to several unrelated features: it is **polysemantic**. To read the features we need to unpack them, which is what sparse autoencoders do (episode 44).



▶ 1:48 Many sparse features share few neurons: each neuron becomes polysemantic.

KEY IDEA

Polysemantic neurons are a natural consequence of compressing many sparse features into few dimensions.

Check 3: What it means for real models

Answer all 1 questions before looking at the answers. Pass mark: all correct.

3.1 TRUE OR FALSE

"If a neuron responds to two unrelated concepts, the model must be poorly trained."

True ☐ False ☐

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. The toy model · rewatch ▶ 0:08-0:47

1.1 0.15. $5 \times (1 - 0.97) = 0.15$: most examples have no active feature at all.

2. Sparsity enables superposition · rewatch ▶ 0:47-1:35

2.1 72° ; $\cos 72^\circ \approx 0.31$. $360^\circ / 5 = 72^\circ$.

2.2 Each is non-negative; after the ReLU, a positive reading means one feature and a negative reading the other. It only fails when both are active at once, which is rare when features are sparse. Antipodal pairs are the cheapest superposition: no interference unless both fire.

2.3 Five norms near 1.1 and angles about 72° apart. The same settings as part 1 of the episode's code.

3. What it means for real models · rewatch ▶ 1:35-1:59

3.1 False. Superposition is an efficient solution when features are sparse.